

PYTHON

Python is a high-level, easy-to-read programming language used to solve problems, analyse data, automate tasks, and build applications.

1. Introduction (intro.py)

- `print()` is used to display output
- **Semicolon ;** is optional in Python

Example:

```
• print("Hello, Python")
```

2. Syntax

Syntax refers to the rules of writing Python code correctly.

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Example:

```
• if 5 > 2:
```

```
    print("Correct")
```

INDENTATION:

Indentation means adding spaces at the beginning of a line to define structure.

Example:

- ```
if True:
 print("Yes")
```

(The space before `print()` is the necessary space that is needed in PYTHON, and is said to be indentation.)

## 3. Statement

A statement is a single line of code that tells Python to perform an action.

## Example:

```
print("Hello")
```

(This above line is said to be a statement)

- There are 3 major types of statements:
  - i. SINGLE STATEMENT
  - ii. MULTIPLE STATEMENTS
  - iii. SEMICOLONS (Rarely used)

### i. SINGLE STATEMENT

It refers to one print statement only.

## Example:

- ```
print("Hello, I'm Anika Jana")
```

ii. MULTIPLE STATEMENTS

It refers to multiple print statements at a time.

Example:

- `print("Hello,I'm Anika Jana")`
- `print("I'm 23 years old")`
- `print("Currently pursuing MBA ON BADS")`

iii. SEMICOLONS

It refers that multiple print statements in one line can be printed one at a time.

Example:

- `print("Hello,"); print("I'm Anika);
print("Jana")`

OUTPUT :

```
Hello,  
I'm Anika  
Jana
```

4. COMMENTS

Comments are lines in the code that Python ignores — used to explain the code.

- `print("Hello, I'm Anika Jana") #print("This
will not run")`

(The grey-shaded statement with the hashtag # in front will not run in Python; it will be ignored.) This is known as comments.)

• MULTIPLE COMMENTS

This is used when multiple statements need to be ignored or it is not needed for the code to run.

For this – `"""` is used.

Example:

```
"""
```

```
• print("Hello,I'm Anika Jana")  
• print("I'm 23 years old")  
• print("Currently pursuing MBA ON BADS")
```

```
"""
```

(As we can see from the above example, multiple statements got ignored all together after using `"""` before and after the statements. This is what known as multiple comments.)

5. VARIABLES

A variable is a container used to store data.

Example:

```
x= 10  
y= "MESSI"  
Print(x)  
Print(y)
```

(Here x & y are the variables that hold the values 10 & MESSI, respectively.)

- **Example:**

```
X= 10
```

```
Y= "MESSI"
```

```
print(X)
```

```
print(Y)
```

```
X= 7
```

```
X= "CR7"
```

```
Print(X)
```

```
Print(Y)
```

```
Y= "MESSI is the G.O.A.T")
```

```
Print(Y)
```

OUTPUT:

10

MESSI

CR7

MESSI

MESSI is the G.O.A.T

(Here, 7 gets ignored and doesn't get printed, as when it sees 2 values back to back in the same variable, it

chooses the last value because the first value gets overlapped by the second one.)

i. CASTING

Casting means converting one data type into another data type.

- Integer Casting (**int()**)

Converts data into an integer (whole number).

Example:

```
x = "10"  
print(int(x))
```

OUTPUT:

```
<class 'int'>
```

- String Casting (**str()**)

Converts into text.

Example:

```
x = 100  
print(str(x))
```

OUTPUT:

```
"100"
```

- Float Casting (**float()**)

Converts into decimal number.

Example:

```
x = "10.5"  
print(float(x))
```

OUTPUT:

10.5

Example:

```
x= 10
```

```
y= "MESSI"
```

```
print(type(x))
```

```
print(type(y))
```

OUTPUT:

<class 'int'>

<class 'str'>

ii. Variable Name

- Should start with an Alphabet.

10Messi × Messi10 ✓

- Case sensitive

A= 26 & a= 26 (Both are not same).

- Cannot use PYTHON keywords

if= 10 × myif= 10 ✓

- CAMEL cases:

Example:

```
myVariableName = "MESSI"
```

(The first letter of the first word should start with lower case and the next following words should have its first letter in upper case.)

- PASCAL cases:

Example:

```
MyVariableName = "MESSI"
```

(Every first letter of all the words should start with upper case.)

- SNAKE cases:

Example:

```
my_variable_name = "MESSI"
```


(Underscore[_] should be placed after every word.)

iii. Assign Multiple Value

- It is a way to assign several variables at once instead of writing separate lines.

Example:

```
X,y,z= "Orange" ,"Banana" ,"Cherry"
```

```
Print(x)
```

```
Print(y)
```

```
Print(z)
```

OUTPUT :

Orange

Banana

Cherry

- One Value to Multiple Variables

Example:

```
X=y=z= "Orange"
```

```
Print(x)
```

```
Print(y)
```

```
Print (z)
```

OUTPUT :

Orange

Orange

Orange

- **Unpack a collection**

Unpacking is the process of breaking a collection into individual variables.

Example:

```
Fruits =["APPLE","BANANA","CHERRY"]
```

```
X, Y, Z = Fruits
```

```
print(X)
```

```
print(Y)
```

```
print(Z)
```

OUTPUT :

APPLE

BANANA

CHERRY

iv. Output Variable

An output variable is a variable whose value is displayed as the final result of a program.

Example 1:

```
X= "Lionel"
```

```
Y= "Messi"
```

```
print(X,Y)
```

OUTPUT:

Lionel Messi

```
X= 10
```

```
Y= "MESSI"
```

```
Print(X,Y)
```

OUTPUT:

10 MESSI

```
X= "lionel"
```

```
Y= "Messi"
```

```
print(X+Y)
```

OUTPUT: Lionel Messi

```
X= "10"
```

```
Y= "Messi"
```

```
Print(X+Y)
```

OUTPUT :

10 Messi

```
X= 10
```

```
Y= "Messi"
```

```
print(X+Y)
```

OUTPUT :

(It won't be able to give the output, as 1 variable is carrying an integer and another variable is carrying a string, and it is said to concatenate the 2 different types of values of 2 different variables)

v. Plain Text

Plain text is simple, unformatted text that contains only characters like letters, numbers, and symbols—without any special styling.

Example:

```
print("Anika Jana")
```

The 2 statements below with print statement-
One with “” & another with “. However, both print statements will give the same output.

❖ `print("Anika Jana")`

❖ `print('Anika Jana')`

❖

vi. Plain Number

The 2 statements below with print statement-
One with “” & another with “. However, both print statements will give the same output.

Example:

```
x = 10    # integer (plain number)
y = 5.5   # float (plain number)
```

```
( + , - , / , * )
```

(They will directly do all the work)